

JORDAN NEWMAN

Backend / Infrastructure Engineer | Distributed Systems, Reliability, Networking

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Selected public repositories are available. The most significant engineering repositories are private; access is available upon request.

SUMMARY

Backend and infrastructure engineer specializing in distributed systems, high-performance networking, reliability, automation, and large-scale production operations. Built and operated infrastructure spanning thousands of servers and developed custom runtime, traffic-analysis, DDoS-mitigation, authentication, telemetry, deployment, kernel, and HTTP/CDN software.

TECHNICAL SKILLS

- **Languages:** C, C++, Go, Python, C#, Objective-C, PHP, Perl, JavaScript, Node.js, Shell, Visual Basic
- **Systems:** FreeBSD, Linux, kernel modules, system calls, operating systems, kqueue, sendfile(), GDB, TCP/IP, non-blocking I/O
- **Networking/Security:** HTTP, DNS, CDN architecture, socket programming, DDoS detection/mitigation, packet capture, libpcap, IPFW, authentication, telemetry
- **Cloud/DevOps:** Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, security scanning, deployment automation
- **Databases/Web:** MySQL, MultiValue/Universe, OAuth/OAuth2, SMTP OAuth, React, jQuery, PHP, web services

SELECTED TECHNICAL HIGHLIGHTS

- Built backend and infrastructure systems supporting approximately 3,000-4,000 servers, 10 CDN locations, and more than 65 Gbps of peak traffic.
- Designed and deployed a custom non-blocking HTTP/CDN runtime across approximately 200 FreeBSD servers.
- Used kqueue, sendfile(), SF_NODISKIO, O_NONBLOCK, TCP_NODELAY, TCP_NOPUSH, accept filters, and specialized I/O workers for scalable event-driven operation.
- Automated deployment, provisioning, monitoring, telemetry, authentication, and infrastructure management across production environments.
- Built packet and DNS analysis systems and DDoS defenses used during attacks involving up to 750,000 compromised or misconfigured hosts.

SELECTED SYSTEMS PROJECTS

- **PrimeHTTPD:** Architected and implemented a high-performance, non-blocking, kqueue-based HTTP/CDN server in C for FreeBSD. Used sendfile(), SF_NODISKIO, TCP_NODELAY, TCP_NOPUSH, O_NONBLOCK, accept_filter_http, persistent connections, and dedicated I/O workers.
- **VirtualDir:** Designed and developed a FreeBSD kernel module that intercepted filesystem-related syscalls and transparently remapped paths from configuration, enabling shared physical layouts without chroot jails; included the vdcli runtime management tool.
- **PrimeDump:** Built a libpcap-based real-time traffic analyzer decoding Ethernet, IP, TCP, UDP, and ICMPv6 with ncurses views and IPFW integration.
- **PrimeDNSTop:** Built a DNS traffic monitor parsing UDP/53 and RFC 1035 queries to detect recursion attacks and rank source and domain activity.
- **TAFOS:** Developed an educational x86 operating-system kernel in C and assembly with a custom MBR bootloader, protected mode, IDT, heap allocator, port I/O, VGA output, and GDB support.

PROFESSIONAL EXPERIENCE

Advantive - Senior Development Specialist

Remote / Tampa Bay, FL | July 2022 - Present

- Implemented Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, and security-scanning workflows across engineering teams.
- Served as a member of the Architecture Team, collaborating with fellow architects and the architecture lead to make technical decisions, define engineering standards, guide platform direction, and develop shared libraries, documentation, and AI-assisted engineering workflows.

DDI Systems - Senior Development Specialist

Manalapan, NJ | November 2021 - July 2022

- Modernized legacy VB.NET components in C# and integrated Microsoft Office 365 SMTP OAuth for secure customer-facing workflows.

ISPrime LLC - CEO

Weehawken, NJ | April 2018 - January 2020

- Led datacenter modernization, operational restructuring, and improvements to internally developed monitoring, security, and infrastructure platforms.

DDI Systems - Computer Programmer

Manalapan, NJ | April 2017 - April 2018

- Built ERP integrations and automated order, catalog, image, SDS, SFTP, SellerCloud, Essendant, and MultiValue/Universe data workflows.

Too Much Media LLC - Computer Programmer

Morganville, NJ | January 2016 - April 2017

- Developed real-time browser chat integrated with ticketing systems and implemented OAuth 2.0 authorization for protected sites.

MFCXY, Inc. - Computer Programmer

Chicago, IL | December 2014 - September 2015

- Developed backend and Windows client features and middleware for cross-platform database migration.

AJPM, LLC - Programmer / Owner

New Jersey / Remote | November 2012 - December 2013

- Built automated purchasing, inventory, repair, resale, and financial workflow software.

ISPrime Inc. - CIO / Partner

Weehawken, NJ | March 2001 - December 2014

Infrastructure Leadership

- Architected and operated hosting and CDN infrastructure spanning approximately 3,000-4,000 servers, 10 locations, multiple datacenters, and more than 65 Gbps of peak traffic.

Systems Software Engineering

- Designed and built PrimeHTTPD, a high-performance, non-blocking, kqueue-based HTTP/CDN server in C for FreeBSD; deployed it across approximately 200 servers and supported more than 150,000 concurrent connections.
- Leveraged `sendfile()`, `SF_NODISKIO`, `TCP_NODELAY`, `TCP_NOPUSH`, `O_NONBLOCK`, `accept_filter_http`, persistent connections, and dedicated I/O workers to keep disk operations from blocking the networking event loop.
- Developed production FreeBSD kernel modules including VirtualDir, KeepClean, and StatCache.

Security and Networking

- Built DDoS detection and mitigation systems, packet and DNS analysis tools, monitoring, telemetry, authentication, deployment, and security software used across production infrastructure.

FatWallet.com - Systems Administrator

Wisconsin / Remote | December 2000 - June 2001

- Rebuilt production infrastructure to resolve performance and stability issues without customer-facing downtime and created automated availability and health alerting.

EDUCATION

Brookdale Community College - Computer Science coursework; left during first semester to pursue ISPrime full-time during rapid company growth.